

Influence of training intensity on adaptations in acid/base transport proteins, muscle buffer capacity, and repeated-sprint ability in active men

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Submitted 13 July 2016; accepted in final form 6 October 2016

McGinley C, Bishop DJ. Influence of training intensity on adaptations in acid/base transport proteins, muscle buffer capacity, and repeated-sprint ability in active men. *J Appl Physiol* 121: 1290–1305, 2016. First published October 14, 2016; doi:10.1152/jappphysiol.00630.2016.—This study measured the adaptive response to exercise training for each of the acid-base transport protein families, including providing isoform-specific evidence for the monocarboxylate transporter (MCT)1/4 chaperone protein basigin and for the electrogenic sodium-bicarbonate cotransporter (NBCe)1. We investigated whether 4 wk of work-matched, high-intensity interval training (HIIT), performed either just above the lactate threshold (HIITΔ20; $n = 8$), or close to peak aerobic power (HIITΔ90; $n = 8$), influenced adaptations in acid-base transport protein abundance, non-bicarbonate muscle buffer capacity ($\beta_{m \text{ in vitro}}$), and exercise capacity in active men. Training intensity did not discriminate between adaptations for most proteins measured, with abundance of MCT1, sodium/hydrogen exchanger (NHE) 1, NBCe1, carbonic anhydrase (CA) II, and CAXIV increasing after 4 wk, whereas there was little change in CAIII and CAIV abundance. $\beta_{m \text{ in vitro}}$ also did not change. However, MCT4 protein content only increased for HIITΔ20 [effect size (ES): 1.06, 90% confidence limits $\times / \div 0.77$], whereas basigin protein content only increased for HIITΔ90 (ES: 1.49, $\times / \div 1.42$). Repeated-sprint ability (5×6 -s sprints; 24 s passive rest) improved similarly for both groups. Power at the lactate threshold only improved for HIITΔ20 (ES: 0.49; 90% confidence limits ± 0.38), whereas peak O_2 uptake did not change for either group. Detraining was characterized by the loss of adaptations for all of the proteins measured and for repeated-sprint ability 6 wk after removing the stimulus of HIIT. In conclusion, 4 wk of HIIT induced improvements in each of the acid-base transport protein families, but, remarkably, a 40% difference in training intensity did not discriminate between most adaptations.

pH regulation; lactate transport; intracellular buffering; high-intensity interval training

NEW & NOTEWORTHY

Our comprehensive analysis of the acid-base transport protein families has shown for the first time that basigin, NBCe1, and the carbonic anhydrase isozymes adapt to exercise training. Despite a 40% difference in training intensity between two work-matched high-intensity interval training (HIIT) groups, intensity did not influence most adaptations in acid-base transporters after 4 wk, and muscle buffer capacity did not change at all. The transience of adaptations was apparent from a detraining response 6 wk after stopping HIIT.

SKELETAL MUSCLE PH HOMEOSTASIS is a balance between hydrogen ion (H^+) production, buffering, and transport. With exercise of

increasing intensity, the greater demand for nonmitochondrial ATP turnover subsequently leads to H^+ production exceeding H^+ consumption. For example, following short-duration, repeated-sprint exercise, intracellular pH (pH_i) typically decreases by 0.2–0.3 pH units (12, 13, 27). Repeated-sprint ability (RSA) may, in turn, partly depend on the intracellular buffer capacity of the muscle (β_m) (28). In addition, transmembrane transport of acids (H^+) and bases (bicarbonate, HCO_3^-) across the sarcolemma provide a further crucial component of pH_i regulation (54). During high-intensity exercise, the largest portion of muscle H^+ transport is lactate coupled, provided by the major skeletal muscle isoforms of the lactate/ H^+ symporters (monocarboxylate transporters: MCT1 and MCT4) (54). An estimated one-third of H^+ removal during high-intensity exercise is nonlactate coupled (6, 7), primarily through the sodium/hydrogen exchanger (NHE) and sodium-coupled bicarbonate transporter (NCBT) proteins. There is also increasing in vitro evidence of a functional and perhaps physical interaction between these transport proteins and the carbonic anhydrase (CA) isozymes (22). However, evidence for the importance of the acid-base transport proteins for exercise performance is equivocal, and the response to exercise training of several of these key proteins remains to be investigated.

Of the biochemical assays of β_m , the ratio of the change in muscle lactate concentration ($[La^-]$) to pH_i ($\beta_{m \text{ in vivo}}$) has been investigated several times. This theoretically provides a measure of total intracellular buffering, comprising physicochemical and metabolic/dynamic buffering (51). However, $\beta_{m \text{ in vivo}}$ is a task-specific measure of β_m and is, therefore, not readily comparable across studies. Indeed, for moderate-intensity exercise involving no change in muscle $[La^-]$, $\beta_{m \text{ in vivo}}$ will be zero, which has been argued to render it a conceptually obscure technique (58). Moreover, because the technique assumes a 1:1 stoichiometry between La^- and H^+ , upregulation of nonlactate-coupled H^+ transport and H^+ -equivalent transport, through increased NHE and NCBT protein abundance, respectively, will result in underestimation of β_m (53). In contrast, the titration technique ($\beta_{m \text{ in vitro}}$) provides an absolute measure of nonbicarbonate physicochemical buffering, primarily from inorganic phosphate and the histidyl-imidazole residues of proteins and carnosine. Some major limitations of this technique are that, in breaking cell walls and membranes during homogenization, extracellular proteins are included in the calculation of $\beta_{m \text{ in vitro}}$ (67, 82); disruption of organelles may expose protonatable residues that are unavailable to bind H^+ in the native state, artifactually inflating β_m (62); and variability in the phosphate profile of samples, owing to phosphagen hydrolysis following homogenization (68, 88), will add to the intra-assay variability. Notwithstanding these limitations, $\beta_{m \text{ in vitro}}$ does provide a nontask-specific measure

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of physicochemical βm , potentially allowing comparison across studies.

It appears exercise intensity may be an important influence on adaptations in $\beta m_{in vitro}$. There is typically no improvement when training is performed below the lactate threshold (LT) (26, 37), whereas higher $\beta m_{in vitro}$ has been measured after high-intensity interval training (HIIT) conducted above the LT, at ~80–90% of peak aerobic power (W_{peak}) (25, 26, 95). Conversely, when training is performed at exercise intensities at or above W_{peak} , there is often no improvement in $\beta m_{in vitro}$ (4, 43, 52, 76), with even a decrease being reported following HIIT undertaken at W_{peak} (15). However, no direct comparison has been made between different training intensities above the LT; therefore, more research is needed to investigate the influence of training intensity on improvements in βm .

While MCT1 appears to adapt to a range of training intensities (11, 17, 20, 24, 72, 81), increases in MCT4 protein content have only been described following maximal [i.e., 100% W_{peak} /maximum O_2 uptake ($\dot{V}O_{2max}$)] or supramaximal training (20, 80, 81), and these are typically of a smaller magnitude than for MCT1. However, there are also several high-intensity training studies that have failed to induce changes in either MCT1 or MCT4 (5, 15, 39, 40, 52, 78). In addition, functioning of both MCTs requires their targeting to the membrane by the chaperone protein basigin (also called CD147 or EMMPRIN), with some evidence in cell models that catalytic activity of the MCTs is dependent on basigin (96). Despite its importance, there is currently no protein evidence of basigin in human skeletal muscle, and no research has investigated the basigin response to exercise training. In contrast, there are several studies showing that expression of the known skeletal muscle NHE isoform, NHE1, increases in response to training (52, 56, 72, 87), although there are conflicting studies describing no change in NHE1 protein content following ostensibly similar interventions (5, 39, 40). Also, there are currently no studies on the NCBT response to training in humans, with unclear protein evidence of the specific skeletal muscle isoforms present. Of the NCBTs, the electrogenic sodium-bicarbonate cotransporters (NBCe1 and NBCe2) have been tentatively

identified, but only using generic antibodies (60). Finally, evidence of the CA protein response to training has not been shown for any of the skeletal muscle isozymes.

Therefore, the present study first sought to provide a comprehensive analysis of the acid-base transport protein response to exercise training. Having detected these proteins, it was possible to manipulate the bioenergetic cost of exercise by using different intensities of work-matched HIIT, and thereby test the hypothesis that a large difference in training intensity would affect upregulation of the acid-base transport proteins and intracellular buffers in skeletal muscle. Finally, we investigated if 6 wk after stopping HIIT there would be physiological detraining, characterized by a reduction in acid-base transport protein content and βm , and decreased RSA.

MATERIALS AND METHODS

Participants

Twenty-seven recreationally active men gave written, informed consent to participate in this study. Eleven participants withdrew during the familiarization period or after the pretraining biopsy, stating a lack of time or unwillingness to undergo further biopsies. The 16 participants who commenced training completed the entire study [age: 23 (5) yr; height: 179.8 (8.6) cm; mass: 80.2 (11.9) kg; peak O_2 uptake ($\dot{V}O_{2peak}$): 48.5 (6.2) $ml \cdot min^{-1} \cdot kg^{-1}$; mean (SD)]. Before the training intervention began, weekly training load was 1,683 (1,399) arbitrary units, mean (SD), calculated from session rating of perceived exertion and training duration (34). All procedures were approved by the Victoria University Human Research Ethics Committee.

Experimental Design

The study employed a two-group parallel design (Fig. 1). There were two work-matched HIIT groups, HIIT Δ 20 (exercised just above the LT; $n = 8$) and HIIT Δ 90 (exercised close to W_{peak} power; $n = 8$), who trained for 4 wk at different intensities above the LT (detailed below). Following the initial pretraining tests, participants were ranked according to total work performed during the RSA test and coded consecutively with either an A or a B in an ABBA sequence. Using random generation of binary numbers, type of training was then randomly allocated to group A (HIIT Δ 90) and group B (HIIT Δ 20).

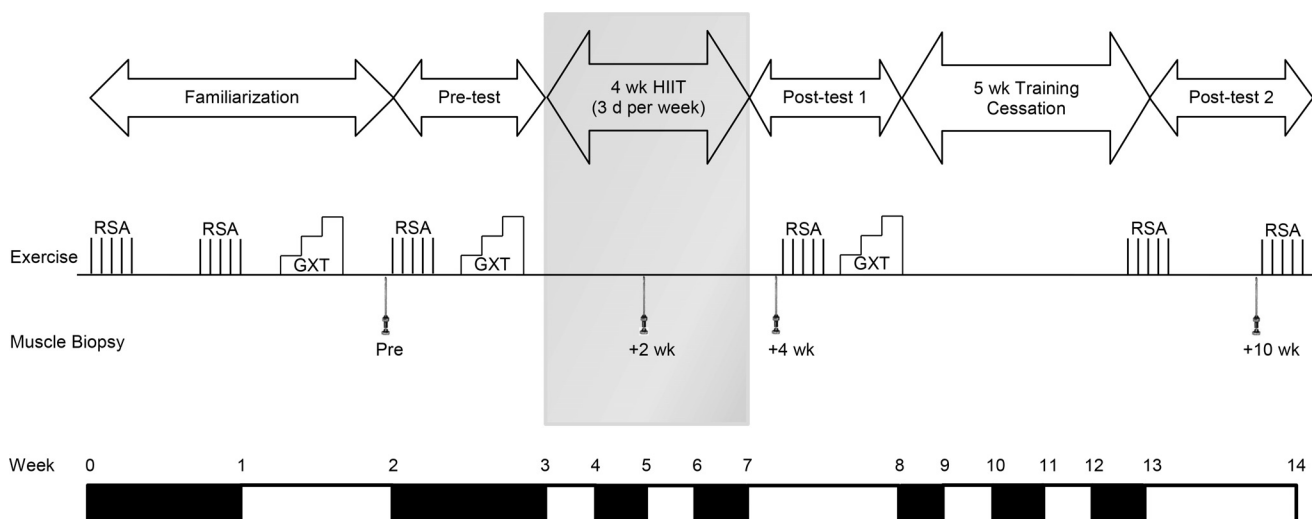


Fig. 1. Experimental design. GXT, graded-exercise test; HIIT, high-intensity interval training; RSA, repeated-sprint ability test.

On separate days before (Pre) and after the 4-wk (+4 wk) training intervention, the following experimental trials were conducted: 1) a resting muscle biopsy, followed 5–10 min later by an RSA test; and 2) a graded-exercise test, followed 5 min later by a $\dot{V}O_{2\text{peak}}$ test. Participants were familiarized with all exercise tests before undertaking experimental trials. Two familiarization trials of the RSA test were conducted on separate days, and on another day familiarization trials of the graded-exercise test and $\dot{V}O_{2\text{peak}}$ test were performed. Muscle biopsies were also taken halfway through the training intervention (+2 wk) and 6 wk after stopping HIIT (+10 wk). To ascertain potential detraining, an RSA test was also performed at +10 wk, again 5–10 min after the muscle biopsy. In the week preceding the +10-wk trial, participants performed a refamiliarization of the RSA test.

HIIT

The training intervention consisted of 4 wk of supervised, work-matched HIIT, performed three times per week on an electromagnetically braked cycle ergometer (Velotron, Racer-Mate, Seattle, WA). The training intensity for both groups was set initially to the power at the LT plus either 20% (HIIT Δ 20) or 90% (HIIT Δ 90) of the difference (Δ) between power at the LT and W_{peak} , i.e.:

$$\text{HIIT}\Delta 20: (LT) + (20\%) \cdot (W_{\text{peak}} - LT)$$

$$\text{HIIT}\Delta 90: (LT) + (90\%) \cdot (W_{\text{peak}} - LT)$$

For the first training session, this was equivalent to 71 (4)% and 98 (1)% of W_{peak} for HIIT Δ 20 and HIIT Δ 90, respectively, similar to training interventions previously published by our group (15, 25, 26, 29). Both groups performed a 5-min steady-state warm-up at 75 W, followed by either 5–15 (HIIT Δ 20) or 4–10 (HIIT Δ 90) 2-min work intervals, interspersed with 1 min of passive recovery (Table 1). Training intensity increased by 5% for the fifth and eighth sessions to maintain the relative training stimulus. For the final training session, all participants performed a standardized bout at the same relative intensity for characterization of short-term postexercise responses for a separate study (70).

Experimental Trials

Participants refrained from exercising and drinking alcohol 24 h before each of the trial days. All trials were performed in the morning (6:30–11:30) following an overnight fast, with participants recording and subsequently replicating a food diary for the 24 h before each exercise trial. The participants were asked to maintain their normal level of activity throughout the study and completed a bespoke web-based daily training diary to monitor compliance.

Needle Muscle Biopsies

Muscle biopsies were taken from the belly of the vastus lateralis using the needle biopsy technique modified with suction (31). Subsequent samples were taken ~1 cm from a previous biopsy site. An incision was made under local anesthesia (1% Xylocaine), and a muscle sample was taken using a Bergström needle (10). Samples were blotted on filter paper to remove blood and then immediately snap-frozen in liquid nitrogen and stored at -80°C until subsequent analyses. Muscle samples were taken from the nondominant leg pretraining (Pre), 2 or 3 days after the sixth (immediately before the seventh) training session (+2 wk), 3 days after the final training session (+4 wk), and 6 wk (+10 wk) after the final training session.

For determination of potential detraining 6 wk after stopping HIIT, failure to adequately maintain a training diary necessitated exclusion of a participant's +10-wk data. One participant was unavailable to undergo the +10-wk biopsy, and a further two participants failed to adequately maintain a training diary; hence, $n = 6$ and $n = 7$ for HIIT Δ 20 and HIIT Δ 90, respectively, at this time point.

RSA Test

The RSA tests were performed seated on an electromagnetically braked cycle ergometer (Excalibur Sport, Lode, Groningen, The Netherlands) and followed a modified protocol of that previously reported (16), comprising 5 $\times$ 6-s maximal sprints, separated by 24 s of rest. Modification entailed using an isokinetic (115 rpm) protocol to negate the effects of cadence on power output and thereby isolate the fatigue-related power component (35). Five seconds before the start of each sprint, participants assumed the ready position: crank of the dominant leg at an angle of 45° . Initially, participants performed a 5-min steady-state warm-up at 75 W, followed by two 3-s practice sprints at 90% of their perceived maximal effort, separated by 24-s rest. A single criterion 6-s maximal sprint was then performed 90 s later, followed by 5 min of rest before the RSA test. To minimize pacing, participants were required to achieve $\geq 90\%$ of their criterion score in the first sprint of the 5 $\times$ 6-s test. If this criterion was not achieved, participants rested for a further 5 min before restarting the test. Consistent verbal encouragement was given throughout. Total work and work decrement were calculated from the raw data, as has been previously reported (16):

$$\text{work decrement (\%)} = \left(\frac{\text{total work} - \text{ideal work}}{\text{ideal work}} \times 100 \right)$$

where *total work* is actual work performed in all 5 sprints, and *ideal work* is work performed in sprint number 1 $\times$ 5.

Table 1. High-intensity interval training programs performed by the HIIT Δ 20 and HIIT Δ 90 training groups

Week	Session	HIIT Δ 20 Group			HIIT Δ 90 Group		
		No. Intervals	Relative Intensity, %	External Power, W	No. Intervals	Relative Intensity, %	External Power, W
1	1	5–6	20 (1)	194 (18)	4	90 (1)	280 (53)
	2	7–9	20 (1)	194 (18)	6	90 (4)	278 (54)
	3	9–10	20 (1)	194 (18)	7	90 (2)	279 (53)
2	4	10–12	20 (1)	194 (18)	8	90 (5)	277 (55)
	5	7–9	30 (2)	204 (19)	6	110 (4)	293 (56)
	6	10–12	30 (2)	204 (19)	8	110 (5)	292 (57)
3	7*	12–15	30 (2)	204 (18)	10	100 (3)	288 (54)
	8	7–9	40 (5)	214 (20)	6	120 (5)	302 (56)
	9	9–10	40 (5)	214 (20)	7	120 (6)	302 (55)
4	10	11–13	40 (6)	218 (19)	9	110 (9)	295 (47)
	11	9–10	40 (5)	214 (20)	7	110 (8)	301 (54)
	12	7	40 (4)	215 (19)	7	40 (3)	229 (54)

Relative intensity was calculated as power at the lactate threshold (LT) plus $x\%$ of the difference (Δ) between power at the LT and peak aerobic power (W_{peak}) [$(LT) + (x\%)(W_{\text{peak}} - LT)$]. Values are means (SD). *Muscle biopsy was performed 2 or 3 days after training session 6, immediately before session 7.

Graded-Exercise Test

Graded-exercise tests were performed pre- and posttraining to determine the LT, from venous blood samples, and W_{peak} . A 20-g intravenous cannula was inserted into an antecubital vein, and samples were taken at rest and at the end of every stage during the graded-exercise test. Before each sample, 1–2 ml of blood were drawn into a dry syringe and discarded. Samples were then drawn into a 3-ml dry syringe and aliquoted into a microtube for immediate analysis of lactate (2300 STAT Plus, YSI, Yellow Springs, OH). The cannula was then flushed with a small volume of sterile saline. The coefficient of variation (CV) for duplicate blood lactate measurements was 4.6%.

Tests were performed on the same cycle ergometer as for the RSA tests, using an intermittent protocol, with 4-min exercise stages and 30-s rest stages. The starting power (90–150 W) was ascertained from the familiarization graded-exercise test, with the goal of minimizing the number of stages to a maximum of 10, and the power was subsequently increased by 30 W for every stage. Participants were required to maintain a set cadence of 70 rpm, and consistent verbal encouragement was provided for the latter stages. The test was terminated either volitionally by the participant, or by the assessors if the participant could no longer maintain the required cadence (± 10 rpm), despite strong verbal encouragement. The LT was identified as the power at which venous blood lactate increased 1 mM above baseline and was calculated using Lactate-E software (77). W_{peak} was calculated as previously reported (61):

$$W_{\text{peak}} = W_{\text{final}} + \left(\frac{t}{240} \times 30 \right)$$

where W_{final} is the power output of the last completed stage, and t is the time in seconds of any final incomplete stage.

$\dot{V}O_{2\text{peak}}$ Test

After the graded-exercise tests, participants performed 5 min of active recovery at 20 W on the cycle ergometer, followed by a square-wave $\dot{V}O_{2\text{peak}}$ test. This comprised a steady-state cycle to volitional fatigue at a supramaximal power output, equating to 105% of W_{peak} achieved during the graded-exercise test. Participants were advised to accelerate to 90–100 rpm at the commencement of a 5-s countdown, and to maintain a high, but not fixed cadence until volitional fatigue. Consistent verbal encouragement was provided throughout. Expired gases were analyzed every 15 s using a custom-made metabolic cart. A two-point calibration of the gas analyzers (S-31A/II and CD-3A analyzers, Ametek) was performed before each test using one certified gravimetric gas (16.1% O_2 , 4.17% CO_2 ; BOC Gases, Chatswood, Australia) and ambient air. Ventilation was recorded every 15 s. The ventilometer (KL Engineering, Sunnyvale, CA) was calibrated at the start of each day using a 3-liter syringe (MedGraphics, St. Paul, MN). $\dot{V}O_{2\text{peak}}$ was calculated as the mean of the two highest consecutive 15-s values. Exercise duration was 138 (36) s [mean (SD)].

Muscle Analyses: βm

Nonbicarbonate βm ($\beta m_{\text{in vitro}}$) was measured on 2–3 mg dry mass (dm) of freeze-dried muscle samples using the titration technique (67). Samples were dissected free of visible blood and connective tissue, then homogenized (Kontes Pellet Pestle, Kimble Chase) on ice for 3×30 s in a 10 mM solution of the glycolytic inhibitor sodium fluoride (0.1 ml NaF/3 mg dm). pH measurements were performed at 37°C with a microelectrode (MI-410, Microelectrodes, Bedford, NH) connected to a pH meter (Laboratory 850, Schott Instruments, Mainz, Germany). After the initial pH measurement, homogenates were adjusted to pH 7.1–7.2, if necessary, with 0.02 M NaOH, and then titrated to pH 6.1–6.2 with the serial addition of 2- μ l aliquots of 0.01 M HCl. From linear regression, the number of moles of H^+ required to change pH from 7.1 to 6.5 was interpolated and then expressed per

whole pH unit and relative to sample dm. Titrations were performed up to five times per sample, depending on available tissue. The typical error (within-sample standard deviation) for repeat titrations was 10.7 mmol $H^+ \cdot kg \text{ dm}^{-1} \cdot pH^{-1}$, equating to an intrasample CV of 7.7%.

Quantitative Western blotting: muscle homogenate preparation. Approximately 30 mg of frozen muscle tissue were homogenized (Kontes Pellet Pestle, Kimble Chase) in a 1:20 dilution of ice-cold RIPA buffer (pH 7.4) containing: 0.15 M NaCl, 1% Triton-X100, 0.5% sodium deoxycholate, 0.05 M Tris, 0.1% SDS, 0.1 M EDTA, 1 μ g/ml aprotinin, 1 μ g/ml leupeptin, 1 mM benzamide, 1 mM sodium orthovanadate, 1 mM NaF, 5 mM sodium pyrophosphate, 1 mM DTT, and 1 mM PMSF. Homogenates were rotated end over end for 60 min at 4°C and centrifuged twice at 15,000g for 10 min at 4°C. The supernatant was then collected, and the pellet discarded.

NBCe1 was assayed in muscle tissue homogenized in a HEPES-sucrose buffer (pH 7.4) containing: 210 mM sucrose, 30 mM HEPES, 40 mM NaCl, 2 mM EGTA, 5 mM EDTA, 1% protease inhibitor cocktail (P8340, Sigma-Aldrich, St. Louis, MI), 1% phosphatase inhibitor cocktail (P5726, Sigma-Aldrich), 1 mM PMSF. Homogenates were prepared exactly as per the RIPA-buffered samples. Owing to insufficient muscle masses, there were lower sample sizes for NBCe1 analysis.

Protein assay. Protein content of muscle homogenate was measured in triplicate using a Bradford assay (Bio-Rad protein assay dye reagent concentrate, Bio-Rad Laboratories, Hercules, CA) against serial dilutions of bovine serum albumin (A9647, Sigma-Aldrich) standards. The CV for the protein assay was 1.9% for RIPA-buffered homogenate and 1.6% for HEPES-sucrose buffered homogenate.

Immunoblotting. RIPA-buffered homogenate was diluted 1:1 in 2× Laemmli buffer (0.125 M Tris, 4% SDS, 20% glycerol, 0.015% bromophenol blue, 10% 2-mercaptoethanol), and equal amounts of total protein (1–25 μ g) were loaded in different wells on 10% or 12% Tris-glycine-HCl SDS-PAGE gels. CAIII proved to be highly abundant, with signal saturation on loading 5 μ g. Homogenate was therefore diluted 1:8 in RIPA buffer (including inhibitors), and 1 μ g was loaded to detect this protein. For CAIV detection, homogenate was loaded on Criterion 4–12% Bis-Tris SDS-PAGE gels (Bio-Rad). For NBCe1, HEPES-sucrose buffered homogenate was diluted 3:1 in 4× Laemmli buffer, and 20 μ g of protein were loaded per well on Criterion 4–12% Bis-Tris SDS-PAGE gels. All samples for a participant were loaded in adjacent lanes on the same gel. Four to six different dilutions of a mixed-homogenate internal standard were also loaded on each gel, and a calibration curve was plotted of density against protein content. From the subsequent linear regression equation, protein abundance was calculated from the measured band intensity for each sample on the gel (73). Coomassie blue (Phastgel Blue R-350, GE Healthcare, Rydalmere, Australia) staining of total protein was used as a loading control (94).

Gel electrophoresis ran for 70–90 min at 140 V for 10% gels and 4–12% gels, or 120–140 min at 100 V for 12% gels. Proteins were wet-transferred to a 0.2- μ m polyvinylidene fluoride membrane at 100 V. Membranes were blocked for 60 min at room temperature in 5% nonfat dry milk (NFDM) diluted in Tris-buffered saline with 0.1% Tween 20 (TBST). Membranes were then washed in TBST and incubated, either at room temperature for 2 h, or overnight at 4°C, with the appropriate primary antibody diluted in 5% bovine serum albumin and 0.02% sodium azide in TBST, or 5% NFDM in TBST for NBCe1 (see Table 2 for antibody conditions). Following TBST washes, the membranes were incubated in the relevant secondary antibody, diluted in 5% NFDM in TBST, for 90 min at room temperature.

After further washes, membranes were incubated in chemiluminescent solution (1.25 mM luminol, 0.2 mM p -coumaric acid, 100 mM Tris pH 8.5, 0.009% H_2O_2) for 2 min, and images were taken with a VersaDoc Imaging System fitted with a charge-coupled device camera (Bio-Rad). Densitometry was performed with Image Laboratory 5.0 software (Bio-Rad) using the volume calculation (sum of all band

Table 2. Details of primary and secondary antibodies used for Western blotting

Antibody	Supplier	Catalog no./Lot no.	Primary Concn.	Secondary Concn.
<i>Primary</i>				
Rabbit polyclonal anti-MCT1	Merck Millipore	AB3540P/2136555	1:1,000	1:15,000
Rabbit polyclonal anti-MCT4	Merck Millipore	AB3316P/2397059	1:1,000	1:20,000
Mouse monoclonal anti-basigin	Santa Cruz	sc-21746/K1913	1:200	1:7,500
Mouse monoclonal anti-NHE1	Merck Millipore	MAB3140/2283852	1:500	1:7,500
Rabbit polyclonal anti-NBCe1	Cell Signaling	11867/0001	1:500	1:5,000
Rabbit polyclonal anti-CAII	Santa Cruz	sc-25596/F0611	1:1,250	1:30,000
Mouse monoclonal anti-CAIII	Abnova	H00000761-M02/12243-S1	1:2,500	1:40,000
Mouse polyclonal anti-CAIV	Abnova	H00000762-B02P/08325 WULz	1:500	1:10,000
Mouse polyclonal anti-CAXIV	Abnova	H00023632-B01P/08358 WULz	1:750	1:10,000
<i>Secondary</i>				
Goat anti-mouse IgG	Perkin Elmer	NEF822001EA		
Goat anti-rabbit IgG	Perkin Elmer	NEF812001EA		

intensities). Background correction was applied individually to each lane using a rolling-ball algorithm, with contrast and brightness adjustments applied homogeneously to the entire image (36). Images are typically displayed with at least five bandwidths above and below the band of interest (2). CVs of replicate gels for individual proteins are presented in their respective figure captions.

Statistical Analyses

To detect a moderate difference in β_m ($f = 0.4$) using a mixed-model ANOVA at $\alpha = 0.05$ and $\beta = 0.2$ required $n = 8$ for each group. Measures of centrality and dispersion are means and standard deviations [mean (SD)], unless otherwise stated. To reduce bias from nonuniformity of error, data were log-transformed where heteroscedasticity was present (75). For these data, geometric mean and standard deviation are reported (geometric mean $\times / \div$ SD). Data were analyzed using linear mixed models with, for example, "time" (repeated-measures), "group", and "group $\times$ time" as fixed factors, and "subject" and "intercept" as random factors. First-order autoregressive covariance structures were used for all models, and model fit was assessed by -2 log-likelihood (33). Effect sizes (ES) were assessed using Cohen's d , where ES thresholds were qualified as trivial < 0.2 , small < 0.6 , moderate < 1.2 , large < 2.0 , very large < 4.0 , and extremely large ≥ 4.0 (49). Uncertainty of effects are expressed as 90% confidence limits (90% CL) and precise P values, unless $P < 0.001$ (21). Effects were not considered meaningful if there was $< 75\%$ probability of being substantially positive relative to the smallest worthwhile change (ES = 0.2) and were deemed unclear if there was a $> 5\%$ probability of being substantially negative (48, 49). If there were no between-group differences, pooled within-group effects are reported. Linear models were analyzed using IBM SPSS Statistics V21 (IBM, Somers, NY), and ES and confidence limits were calculated using custom Excel spreadsheets (48).

RESULTS

Training Data

Both training groups performed similar volumes of total work throughout the training intervention, 2.74 (0.23) MJ and 2.88 (0.57) MJ for HIIT Δ 20 and HIIT Δ 90, respectively. There was excellent adherence to the training program, with one participant in each group missing a single training session due to illness. The different metabolic costs for the two training intensities are shown by mean heart rate responses for the penultimate training session (Fig. 2).

Protein Abundance

Basigin (CD147/EMMPRIN). The predicted molecular mass for basigin-2, the canonical basigin isoform and probable

skeletal muscle isoform (64), is 42 kDa. Basigin has three *N*-glycosylation sites and functions in both a high- and a less-glycosylated form, giving a range of apparent molecular masses (89). Using a monoclonal anti-basigin antibody (sc-21746, Santa Cruz Biotechnology, Santa Cruz, CA), a single band was detected just above 37 kDa in muscle homogenate (Fig. 3) and at ~ 50 kDa in a Hep G2 cell lysate positive control (sc-2227, Santa Cruz). To confirm the presence of basigin, duplicate gels were run in parallel, same homogenate and same running and transfer tanks, and probed with the first monoclonal anti-basigin antibody and a second monoclonal anti-basigin antibody (sc-46700, Santa Cruz). This second antibody had a poorer signal-to-noise ratio and detected a 50-kDa band and several weaker lower molecular mass bands, including just above 37 kDa (Fig. 3), but did not detect any signal with the Hep G2 cell lysate. For additional confirmation, blots were reprobbed using a different species (goat) polyclonal anti-basigin antibody (sc-9757, Santa Cruz). This was performed following a 15-min incubation with H_2O_2 at $37^\circ C$ to inactivate the peroxidase of the horseradish peroxidase-conjugated secondary antibody (86). Again, the antibody produced weak signal, but did detect the > 37 -kDa band in muscle homogenate and ~ 50 -kDa band in Hep G2 cell lysate. With this supporting data, the single > 37 -kDa band detected by the first mouse monoclonal antibody (sc-21746) was quantified.

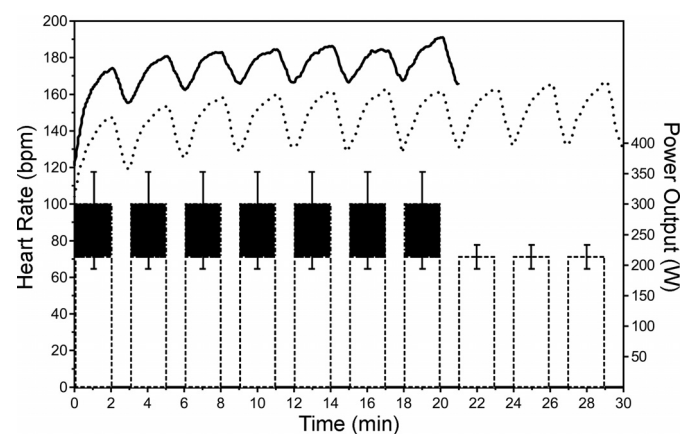


Fig. 2. Mean heart rate during training session 11 for the HIIT Δ 20 ($n = 8$, dotted line) and HIIT Δ 90 ($n = 7$, solid line) groups. Bars show the external power [mean (SD)] for HIIT Δ 20 (open) and HIIT Δ 90 (solid). Exercise comprised 2-min intervals interspersed with 1 min of passive rest.

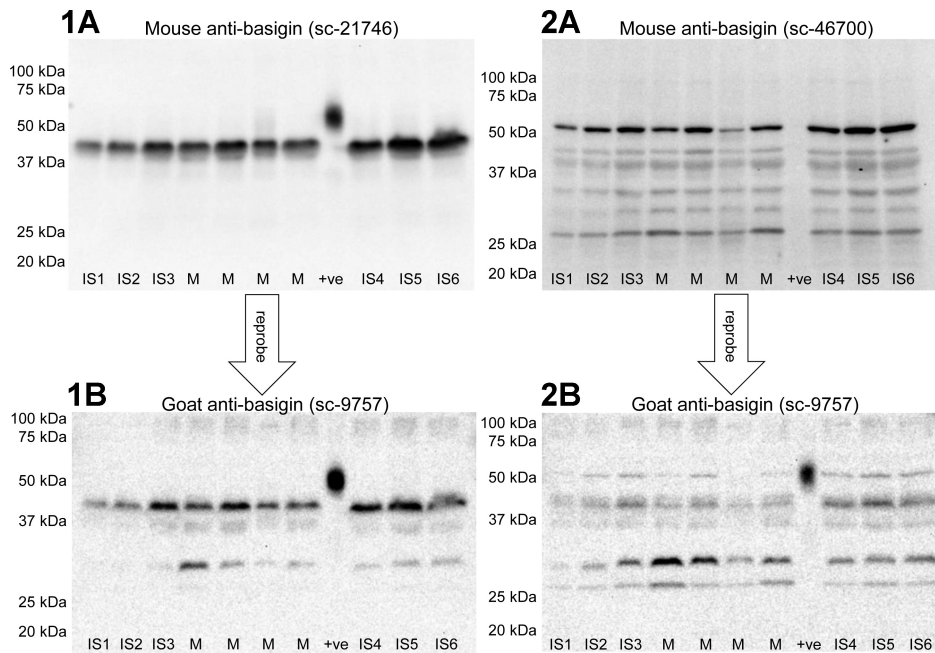


Fig. 3. Basigin antibody validation. Using Western blotting, two duplicate gels (1 and 2) were loaded with four lanes of a single sample of skeletal muscle homogenate (M), six different concentrations of an internal standard (IS1–IS6: 8–38 μ g protein), and one lane of a Hep G2 cell lysate positive control (+ve). Following transfer, *membrane 1A* was immunoblotted with a mouse monoclonal anti-basigin antibody (sc-21746), and *membrane 2A* was immunoblotted with a different mouse monoclonal anti-basigin antibody (sc-46700). After imaging, the peroxidase of the horseradish peroxidase-conjugated secondary antibodies was inactivated by 15 min of H_2O_2 incubation. Both membranes were reprobbed (*1B* and *2B*) with a goat polyclonal anti-basigin antibody (sc-9757). See text for additional details on immunoblotting.

Overall basigin abundance changed in response to training (time main effect: $F_{3,37.2} = 4.47$, $P = 0.009$) (Fig. 4C). Relative to Pre, basigin increased more for HIIT Δ 90 than HIIT Δ 20 at +4 wk (ES: 1.49, 90% CL $\times / \div$ 1.42), with no clear differences at +2 wk (ES: 0.64, $\times / \div$ 0.95) or +10 wk (ES: 0.69, $\times / \div$ 1.06) (group $\times$ time interaction: $F_{3,37.2} = 1.78$, $P = 0.17$).

MCT1. The MCT1 antibody recognized a single band at ~50 kDa (Fig. 4A). MCT1 protein content increased for both groups (time main effect: $F_{3,40.2} = 13.00$, $P < 0.001$), with no difference between the two groups (group $\times$ time interaction: $F_{3,40.2} = 0.08$, $P = 0.97$, +4 wk ES: 0.25, $\times / \div$ 1.75). On pooling the data for the two groups, there was a moderate increase at +2 wk (ES: 0.60, $\times / \div$ 0.60) and a very large increase at +4 wk (ES: 2.49, $\times / \div$ 0.83). MCT1 abundance decreased 6 wk after stopping HIIT but remained elevated above baseline (+10 wk ES: 0.90, $\times / \div$ 0.59).

MCT4. The MCT4 antibody recognized a strong band at ~50 kDa and a weaker band at ~75 kDa in some samples (Fig. 4B); only the 50-kDa band was quantified. Increases in MCT4 protein content were only evident for HIIT Δ 20 (group $\times$ time interaction: $F_{3,42.1} = 4.5$, $P = 0.008$). There was no difference between groups at +2 wk (ES: 0.22, $\times / \div$ 0.70), but at +4 wk MCT4 had increased for HIIT Δ 20 relative to HIIT Δ 90 (ES: 1.06, $\times / \div$ 0.77). The greater MCT4 abundance was lost 6 wk after stopping HIIT, with no clear difference between groups at +10 wk (ES: 0.30, $\times / \div$ 0.59).

NHE1. The NHE1 antibody recognized a single or a double band just below 100 kDa (Fig. 4D); both bands were quantified if present (57). There was an increase in NHE1 protein content for both groups (time main effect: $F_{3,44.2} = 11.35$, $P < 0.001$), with no difference between the two groups (group $\times$ time interaction: $F_{3,44.2} = 0.45$, $P = 0.72$, +4 wk ES: 0.03, $\times / \div$ 0.88). On pooling the data, compared with Pre, there was no difference at +2 wk (ES: 0.23, $\times / \div$ 0.44), but a large increase was evident at +4 wk (ES: 1.34, $\times / \div$ 0.41). There was still a clear effect at +10 wk (pooled ES: 0.58, $\times / \div$ 0.45), indi-

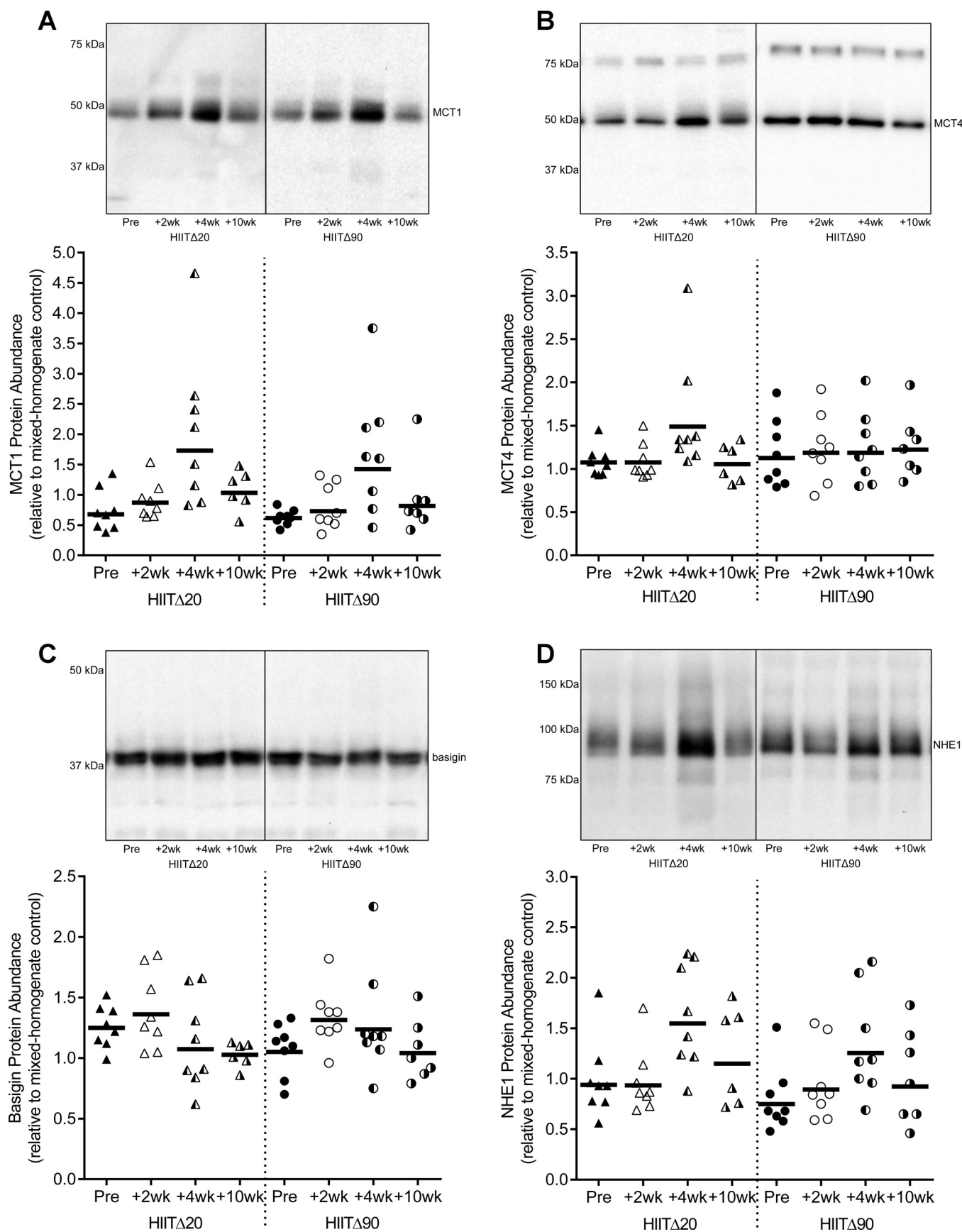
cating that NHE1 abundance had not yet returned to baseline values.

NBCe1. The NBCe1 antibody was raised against a COOH-terminus sequence and recognized a band close to the predicted molecular mass of 121 kDa for the canonical B isoform (Fig. 5A). Stronger signal bands were also detected at lower molecular masses of ~50 kDa and 75 kDa. It is unknown if these bands represent as yet unidentified solute carrier 4 proteins, or possibly proteolytic fragments of NBCe1, as has been suggested previously with other NBC antibodies (84). In the absence of additional confirmation, only the ~120-kDa band was quantified. To avoid signal saturation from the lower molecular mass bands affecting quantification, membranes were cut at 75 kDa.

No clear differences were found in NBCe1 protein content between the groups (group $\times$ time interaction: $F_{3,16.7} = 0.82$, $P = 0.50$, +4 wk ES: 0.49, $\times / \div$ 1.40). NBCe1 increased in response to training (time main effect: $F_{3,16.7} = 12.3$, $P < 0.001$), with no change at +2 wk (pooled ES: 0.20, $\times / \div$ 0.55), but increased abundance at +4 wk (pooled ES: 1.31, $\times / \div$ 0.67), compared with Pre.

CAII. The CAII antibody recognized a single band just above 25 kDa (predicted molecular mass 29 kDa) (Fig. 6A). CAII protein content did not differ between the two groups (group $\times$ time interaction: $F_{3,37.7} = 0.54$, $P = 0.66$, +4 wk ES: 0.15, $\times / \div$ 1.68). Overall, CAII increased after 4 wk of training (time main effect: $F_{3,37.7} = 7.21$, $P = 0.001$, pooled ES: 1.35, $\times / \div$ 0.80), but did not change after 2 wk (pooled ES: -0.09, $\times / \div$ 0.51). CAII abundance decreased back to baseline 6 wk after stopping HIIT (pooled +10 wk ES: 0.15, $\times / \div$ 0.46).

CAIII. The CAIII antibody recognized a single band just above 25 kDa (predicted molecular mass 30 kDa) (Fig. 6B). There was no overall training effect for CAIII abundance (time main effect: $F_{3,37.6} = 0.46$, $P = 0.71$, pooled +4 wk ES: 0.16, $\times / \div$ 0.51), but the two groups did respond differently to training (group $\times$ time interaction: $F_{3,37.6} = 3.03$, $P =$



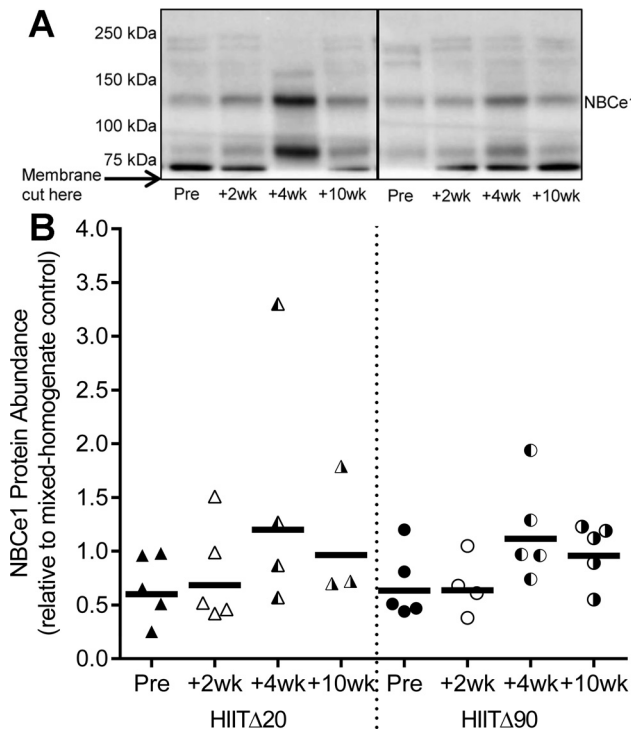


Fig. 5. Representative immunoblot (A) and NBCe1 protein abundance (B) before (Pre) and after 2 wk (+2 wk) and 4 wk (+4 wk) of high-intensity interval training (HIIT), and 6 wk after stopping HIIT (+10 wk), for the HIIT Δ 20 and HIIT Δ 90 training groups. Protein abundance is relative to a calibration curve of a mixed-homogenate internal standard run on each gel. Individual data points and geometric means (horizontal bars) are plotted. Nonadjacent lanes from the same blot are indicated by a vertical line. CV of fold-change data for duplicate gels was 24%.

0.04). The difference between groups was because of a small reduction in CAIII abundance at +2 wk for HIIT Δ 90 of 0.86-fold $\times / \div 1.28$ and a small increase for HIIT Δ 20 of 1.09-fold $\times / \div 1.20$ (ES: 1.01, $\times / \div 0.84$), compared with Pre. At +4 wk and +10 wk there was no difference between groups (+4 wk ES: -0.48 , $\times / \div 1.05$, +10 wk ES: -0.07 , $\times / \div 0.72$).

CAIV. There was low abundance of CAIV and, consequently, a poor signal-to-noise ratio. Multiple bands were detected, but the antibody recognized a single or double band at the predicted molecular mass of 35 kDa in the muscle homogenate and positive control (293T Cell Transient Overexpression Lysate, Abnova, Taipei, Taiwan). Similar to earlier reports (92), a strong band at ~ 75 kDa was recognized in the muscle homogenate but not the positive control. To test whether this band was a CAIV dimer, different concentrations of DTT (0.5 mM to 10 mM) were added to the homogenate in an attempt to break potential disulfide bonds. A similar approach proved successful in increasing CAIX monomer expression at the expense of dimer expression in recombinant proteins (47), but, for the present experiment, additional DTT

had no effect on the 75-kDa band (Fig. 7). Therefore, only the 35-kDa band was quantified (Fig. 6C).

CAIV protein content did not differ between the two groups (group $\times$ time interaction: $F_{3,28.3} = 0.74$, $P = 0.54$, +4 wk ES: 0.20, $\times / \div 0.83$). There were no clear changes in response to training (time main effect: $F_{3,28.3} = 1.72$, $P = 0.19$). On pooling the data, protein content decreased at +2 wk (ES: -0.30 , $\times / \div 0.25$), but was no different to baseline at +4 wk (ES: 0.16, $\times / \div 0.39$) and +10 wk (ES: -0.15 , $\times / \div 0.47$).

CAXIV. The CAXIV antibody recognized a single band below 50 kDa in the muscle homogenate (predicted molecular mass 38 kDa) and a positive control (293T Cell Transient Overexpression Lysate, Abnova) (Fig. 6D). CAXIV abundance did not differ between the two groups (group $\times$ time interaction: $F_{3,41.3} = 0.29$, $P = 0.83$, +4 wk ES: 0.32, $\times / \div 0.62$). On pooling the data, there was greater protein content at +2 wk (ES: 0.59, $\times / \div 0.45$) and at +4 wk (ES: 0.46, $\times / \div 0.30$), compared with Pre (time main effect: $F_{3,41.3} = 12.3$, $P < 0.001$). These adaptations were lost 6 wk after stopping HIIT (+10 wk ES: -0.41 , $\times / \div 0.59$).

βm

Pre $\beta m_{in vitro}$ was 149.1 (11.7) and 138.8 (13.6) mmol $H^+ \cdot kg \text{ dm}^{-1} \cdot pH^{-1}$, and at +4 wk was 144.0 (19.4) and 132.5 (16.4) mmol $H^+ \cdot kg \text{ dm}^{-1} \cdot pH^{-1}$, for HIIT Δ 20 and HIIT Δ 90, respectively (Fig. 8). $\beta m_{in vitro}$ did not change differently between groups (group $\times$ time interaction: $F_{3,44.9} = 0.18$, $P = 0.91$, +4 wk ES: 0.08; 90% CL ± 0.72). Overall, there was a small decrease at +4 wk compared with Pre (time main effect: $F_{3,44.9} = 1.32$, $P = 0.28$, pooled +4 wk ES: -0.40 ; ± 0.34). This decrease of 5.7 mmol $H^+ \cdot kg \text{ dm}^{-1} \cdot pH^{-1}$ was less than the typical error of measurement of 10.7 mmol $H^+ \cdot kg \text{ dm}^{-1} \cdot pH^{-1}$.

RSA

Total work performed during the RSA test did not change in response to training (time main effect: $F_{2,17.8} = 0.97$, $P = 0.40$, +4 wk pooled ES: 0.04; ± 0.19) (Fig. 9A). There were no between-group differences for the change in work decrement in response to training (group $\times$ time interaction: $F_{2,24.5} = 1.10$, $P = 0.36$, +4 wk ES: 0.42; ± 0.81) (Fig. 9B). On pooling the data, both groups improved similarly relative to baseline (time main effect: $F_{2,24.5} = 7.36$, $P = 0.003$, +4 wk ES: 0.59; ± 0.39), with a small improvement in work decrement from -12.2 (4.6)% Pre to -9.3 (3.3)% at +4 wk. This improvement was lost 6 wk after stopping HIIT (ES: 0.15; ± 0.44).

Aerobic Capacity

Power at the LT improved for HIIT Δ 20 from 172 (26) W to 194 (25) W (Fig. 10A), but remained unchanged for HIIT Δ 90 from 189 (52) W to 190 (45) W (group $\times$ time interaction: $F_{1,16} = 5.85$, $P = 0.03$, ES: 0.49; ± 0.38). There was no difference between the two groups for W_{peak} (group $\times$ time

Fig. 4. Representative immunoblots and protein abundance of MCT1 (A), MCT4 (B), basigin (C), and NHE1 (D) before (Pre) and after 2 wk (+2 wk) and 4 wk (+4 wk) of high-intensity interval training (HIIT), and 6 wk after stopping HIIT (+10 wk), for the HIIT Δ 20 and HIIT Δ 90 training groups. Protein abundance is relative to a calibration curve of a mixed-homogenate internal standard run on each gel. Individual data points and geometric means (horizontal bars) are plotted. Separate blots (A, B, and D) or nonadjacent lanes from the same blot (C) are indicated by vertical lines. CVs of fold-change data for two to four repeat gels are as follows: MCT1 (21%), MCT4 (13%), basigin (22%), and NHE1 (21%).

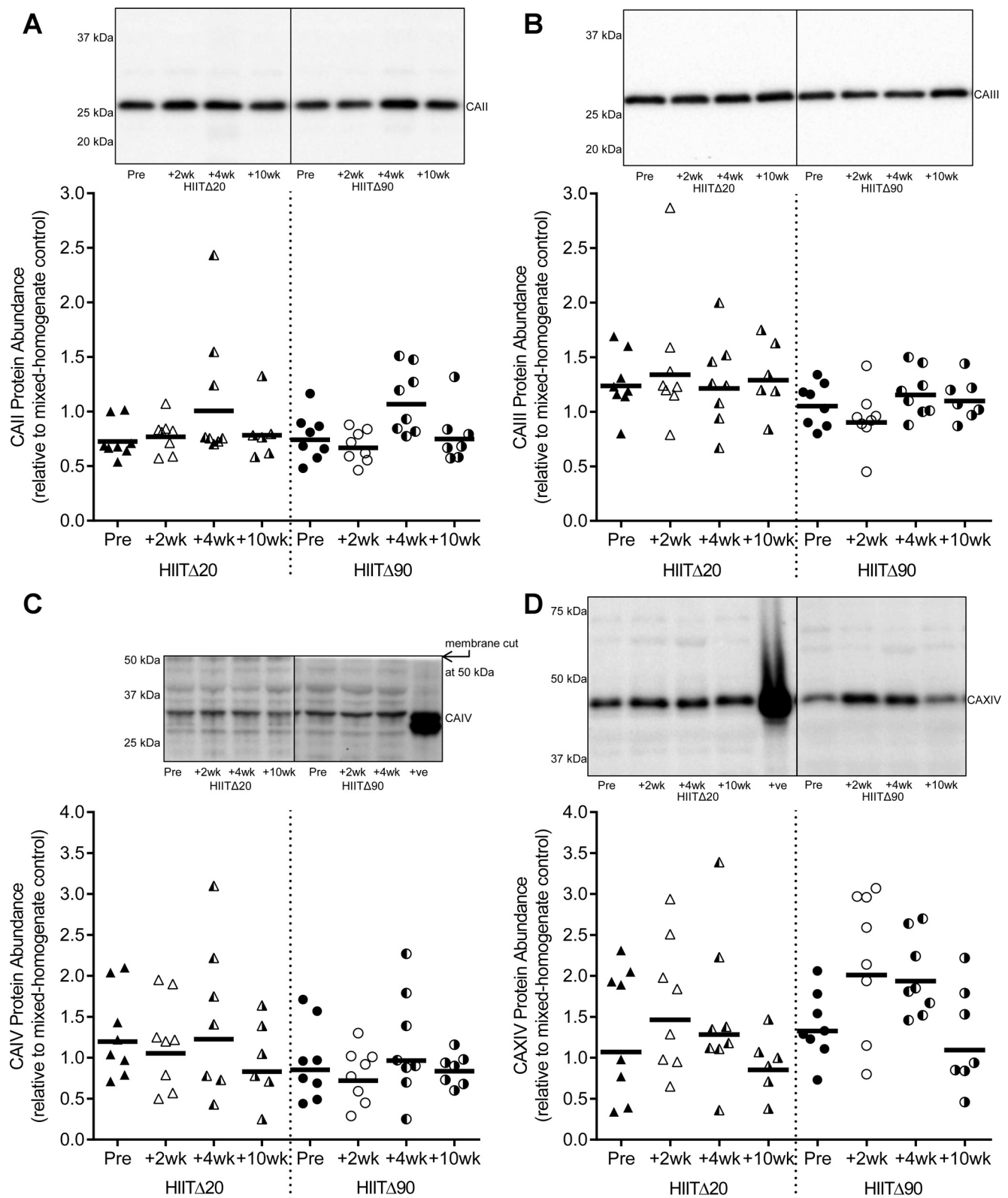


Fig. 6. Representative immunoblots and protein abundance of CAII (A), CAIII (B), CAIV (C), and CAXIV (D) before (Pre) and after 2 wk (+2 wk) and 4 wk (+4 wk) of high-intensity interval training (HIIT), and 6 wk after stopping HIIT (+10 wk), for the HIITΔ20 and HIITΔ90 training groups. Protein abundance is relative to a calibration curve of a mixed-homogenate internal standard run on each gel. Individual data points and geometric means (horizontal bars) are plotted. Positive controls (+ve) were loaded for CAIV and CAXIV. Nonadjacent lanes from the same blots (A and B) and from separate blots (C and D) are indicated by a vertical line. CVs of fold-change data for two or three repeat gels are as follows: CAII (22%), CAIII (single gels), CAIV (single gels), and CAXIV (25%).

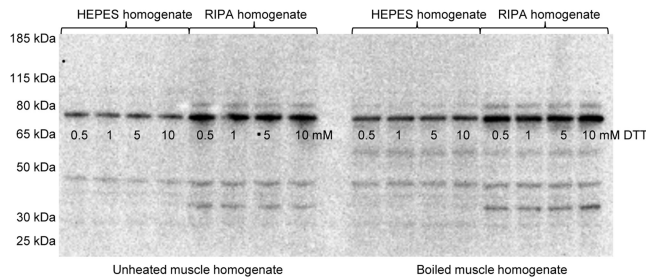


Fig. 7. Effect of adding increasing concentrations of DTT to homogenate on bands detected by the CAIV antibody. To test whether a ~75-kDa band detected was a dimer of CAIV, 0.5–10 mM DTT were added to RIPA or HEPES-sucrose buffered muscle homogenate in an attempt to break potential disulfide bonds. The homogenate was either unheated, or boiled at 95°C for 5 min. Proteins were separated on a 4–12% Bis-Tris SDS-PAGE gel. See text for additional details on immunoblotting.

interaction: $F_{1,16} = 3.7 \times 10^{-4}$, $P = 0.99$, ES: $0; \pm 0.30$). Pre W_{peak} was 272 (16) W and 285 (53) W (Fig. 10B), and at +4 wk W_{peak} was 280 (11) W and 294 (52) W, for HIITΔ20 and HIITΔ90, respectively. Therefore, W_{peak} did not change meaningfully with training (time main effect: $F_{1,16} = 6.51$, $P = 0.02$, pooled ES: $0.20; \pm 0.14$).

Pre $\dot{V}O_{2\text{peak}}$ was 51.0 (5.5) $\text{ml} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$ and 47.7 (5.5) $\text{ml} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$, and at +4 wk $\dot{V}O_{2\text{peak}}$ was 50.1 (3.4) $\text{ml} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$ and 48.5 (5.1) $\text{ml} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$, for HIITΔ20 and HIITΔ90, respectively (Fig. 10C). There was no difference in $\dot{V}O_{2\text{peak}}$ between the two groups (group $\times$ time interaction: $F_{1,15} = 1.06$, $P = 0.32$, ES: $0.27; \pm 0.51$), and no overall change following training (time main effect: $F_{1,15} = 0.007$, $P = 0.93$, pooled ES: $0.00; \pm 0.25$).

DISCUSSION

This study provides a comprehensive investigation of the response to exercise training of the known skeletal muscle acid-base transport proteins. For the first time in humans,

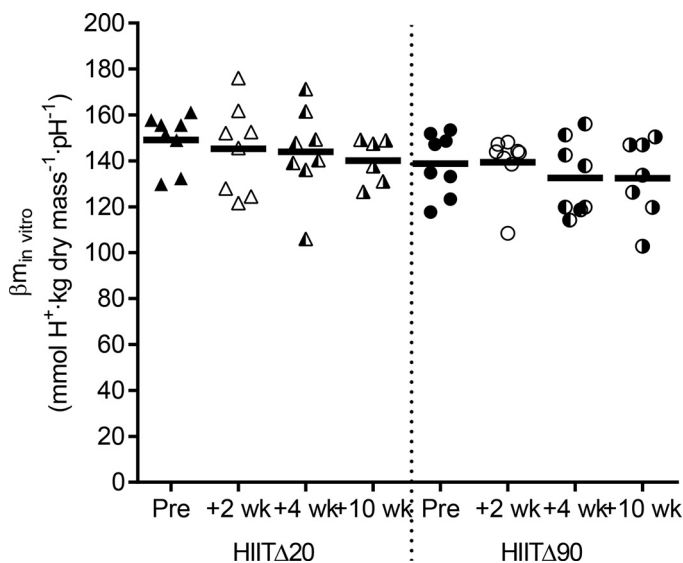


Fig. 8. Nonbicarbonate muscle buffer capacity ($\beta_{\text{m in vitro}}$) before (Pre) and after 2 wk (+2 wk) and 4 wk (+4 wk) of high-intensity interval training (HIIT), and 6 wk after stopping HIIT (+10 wk), for the HIITΔ20 and HIITΔ90 training groups. Individual data points and means (horizontal bars) are plotted. Intrasample CV for 2–5 repeat titrations was 7.7%.

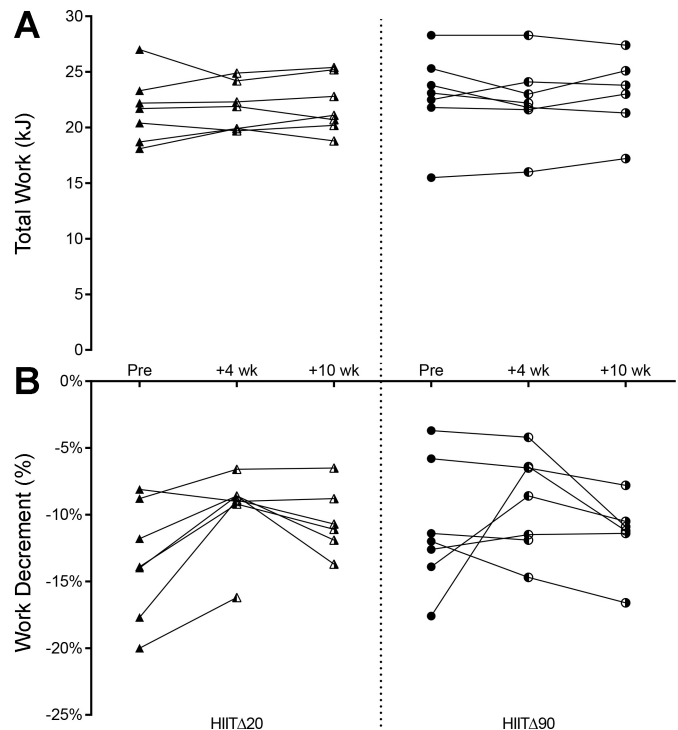


Fig. 9. Total work (A) and work decrement (B) during a repeated-sprint ability (RSA) test before (Pre) and after 4 wk (+4 wk) of high-intensity interval training (HIIT), and 6 wk after stopping HIIT (+10 wk), for the HIITΔ20 and HIITΔ90 training groups. Individual data points are plotted.

training-induced changes have been reported for the MCT chaperone protein basigin, NBCe1, and the four skeletal muscle CA isozymes. The major findings are that the abundance of most of the acid-base transport proteins increased following 4 wk of HIIT. However, despite a 40% difference in the external power at which HIITΔ20 and HIITΔ90 trained, adaptations were typically not different between the two training intensities employed (i.e., 71–79% and 98–106% of W_{peak} , respectively). Regardless of training intensity, six sessions of HIIT performed over the first 2 wk provided insufficient stimulus to provoke upregulation of most of the acid-base transport proteins or $\beta_{\text{m in vitro}}$. Furthermore, the transience of adaptations was apparent from a detraining response 6 wk after removing the physiological stress of HIIT.

As a chaperone protein, basigin not only targets MCT1 and MCT4 to the membrane, but also maintains activity of these transporters (41). Despite its crucial function in La^-/H^+ transport, there are no published reports of basigin protein in human skeletal muscle. Of the four known isoforms, basigin-1 and basigin-4 are retina specific (64), while basigin-2 and basigin-3 have predicted molecular masses of 42 and 19 kDa, respectively. However, with three *N*-glycosylation sites, there are a range of potential apparent molecular masses (64, 89). mRNA data have previously shown the 42-kDa canonical basigin-2 isoform to be the most highly expressed human skeletal muscle isoform (64). To attempt to identify the specific isoform in the present study, duplicate immunoblots were separately probed with two monoclonal mouse anti-basigin antibodies, and then reprobated with a polyclonal goat anti-basigin antibody, producing in all cases the same band above 37 kDa in muscle

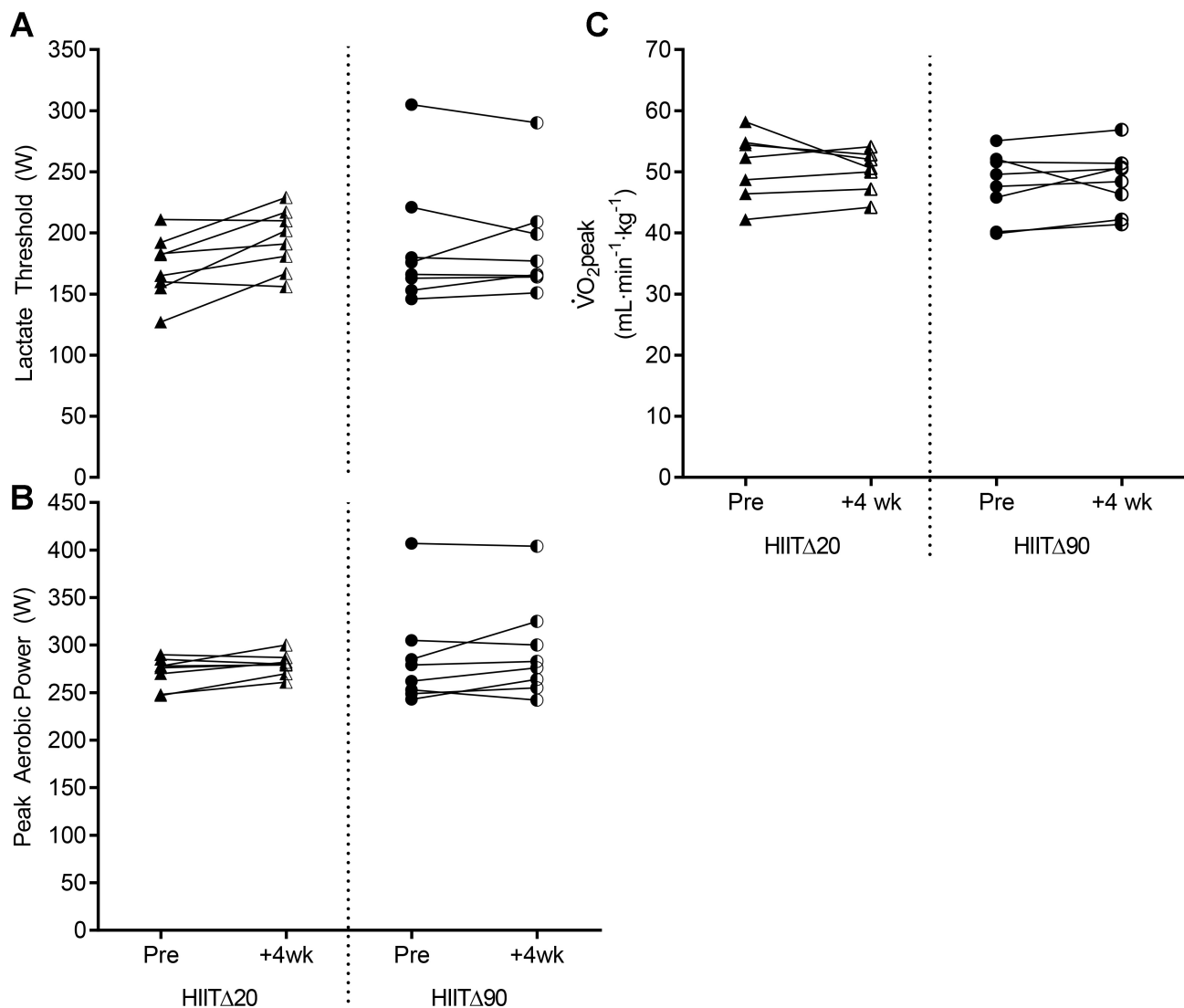


Fig. 10. Lactate threshold (A) and peak aerobic power (B) determined from a graded-exercise test, and peak oxygen uptake ($\dot{V}O_{2peak}$; C) during a square-wave supramaximal test before (Pre) and after 4 wk (+4 wk) of high-intensity interval training (HIIT), for the HIITΔ20 and HIITΔ90 training groups. Individual data points are plotted.

homogenate (Fig. 3). Therefore, it is probable that the basigin-2 isoform was detected here.

Following this validation, basigin has been shown for the first time to adapt to exercise training. Basigin protein abundance increased in response to 4 wk of training for HIITΔ90 (1.18-fold $\times$ / $\div$ 1.40) but not HIITΔ20 (0.86-fold $\times$ / $\div$ 1.41), indicating that training intensity influenced changes. Previous research in L6 cells reported a dose-response upregulation of basigin protein expression following 1 h of incubation with 10 mM and 20 mM of La^- (44). Given it is likely there was higher $[La^-]$ following HIITΔ90 than HIITΔ20, our data support the idea that La^- accumulation may be an important signal for basigin translation, but additional research is needed to test this supposition.

MCT4 increased for HIITΔ20 (1.38-fold $\times$ / $\div$ 1.32) but not HIITΔ90 (1.05-fold $\times$ / $\div$ 1.12) after 4 wk of training. Increased MCT4 protein content has seldom been reported, with a magnitude of less than 1.5-fold in those studies reporting a change (20, 80, 81). Therefore, the present data are

consistent with some (20, 80, 81), but not most (5, 11, 15, 32, 39, 40, 52, 71, 83, 87) modalities of HIIT inducing modest increases in MCT4. In contrast, the large and variable increases in MCT1 abundance for both groups after 4 wk, but not 2 wk, indicates the intensity between the LT and W_{peak} at which HIIT was performed did not influence upregulation. These data support previous research indicating that MCT1 and MCT4 respond to distinct energetic stresses (18, 44), although their respective signaling pathways remain to be described fully.

NHE1 abundance also increased similarly for both groups after 4 wk, but not 2 wk, of training. Although high-intensity training has been shown to be important for the upregulation of NHE1 in humans (40, 52, 87) and rodents (55), the present data suggest that any high-intensity training performed between the LT and W_{peak} may provide sufficient stimulus. However, given that some other studies have found little or no change with 6–9 wk of different modalities of HIIT (5, 39), it is probable that initial training status determines how readily upregulation of NHE1 abundance is

stimulated (5). If high-intensity training is already a regular component of training, further increases in NHE1 protein content may be unlikely in the short term.

This study has provided the first isoform-specific evidence for any of the NCBTs in human skeletal muscle, identifying NBCe1 using an antibody raised against the COOH-terminus of human NBCe1. Previous studies in human skeletal muscle have used a polyclonal antibody raised against a 54 amino acid sequence of the NH₂-terminus of the NBCe1-A isoform (57, 60). Performing a BLAST alignment (99) of this sequence of NBCe1-A and four other putative skeletal muscle NCBTs (NBCe2, NBCn1, NBCn2, NDCBE) produced sequence identities ranging from 49% to 57%. Given that the NCBTs have relatively similar predicted molecular masses between 116 and 140 kDa (79), previous immunoblotting data with this NH₂-terminus antibody could not discriminate between isoforms. In addition, we provide the first evidence of training-induced changes in any of the NCBT proteins in humans, with NBCe1 protein content increasing similarly for both training groups. Despite the high individual variability and low sample sizes, from this preliminary evidence it seems that NBCe1 can be upregulated following HIIT.

For the first time, this study has demonstrated that the CAs can be upregulated following exercise training, although the intensity at which HIIT was performed did not affect adaptations. Despite high individual variability, mean abundance of both CAII and CAXIV was found to increase following 4 wk of HIIT in both groups, but CAIII and CAIV changed little. While the relative importance of these changes in vivo is uncertain, in vitro research suggests there may be enhanced acid-base transport and intra/extracellular buffering (8). The extracellular facing sarcolemmal CAs catalyze buffering of H⁺ in the interstitium by HCO₃⁻ (50), thereby maintaining a pH gradient relative to the intracellular space necessary for high MCT activity. Increasingly evidence is showing that the role of the CA isozymes in pH regulation is not simply limited to a catalytic function in reversible hydration of CO₂. Despite some doubt about whether direct binding of CAs and acid-base transport proteins occur (19), convincing in vitro data using animal and cell models indicate an important functional interaction between CAII, CAIV, and MCT1/4 (9, 42); between NBCe1 and CAII, CAIII, and CAIV (1, 85); and possibly also between CAII, CAIV, and NHE1 (63, 97). Greatest enhancement of transport activity by the CAs would be from increases in abundance of both cytosolic and sarcolemmal isozymes, as has been seen in this study with CAII and CAXIV. The extent to which transport activity is enhanced by CA protein content remains uncertain at this time, and, given that there were no differences between the two training groups, the stimulus for CA upregulation requires further investigation.

It is challenging to make robust inferences on the physiological relevance of our data because transport activity is dependent on multiple factors, not only total abundance of specific transport proteins (66). For example, the direction and magnitude of La⁻/H⁺ transport are dependent on La⁻ and pH gradients, absolute pH, temperature, and Michaelis-Menten (*K_m*) kinetics. MCT1 transport activity is maximal at low [La⁻] due to a *K_m* for L-lactate of ~3–5 mM (i.e., high binding affinity), and MCT4 transport activity progressively increases with increasing [La⁻] due to a *K_m* of ~30 mM (low binding

affinity) (23, 66). MCT1 and MCT4 are primarily involved in La⁻/H⁺ influx and efflux, respectively, with La⁻ efflux occurring at a higher maximal rate than La⁻ influx (93). Therefore, the increase in MCT1 abundance for the HIITΔ20 and HIITΔ90 groups would likely have resulted in higher rates of La⁻ influx, and subsequently higher capacity for La⁻ oxidation (30). With a corresponding increase in NHE1 and NBCe1 abundance for both groups, nonlactate-coupled H⁺ removal from muscle fibers may have increased for both groups (56), whereas increased MCT4 abundance for HIITΔ20 would have enhanced La⁻/H⁺ removal at increasingly higher [La⁻]. Yet both MCTs are also targeted to the membrane by basigin; therefore, the increase in basigin protein content for HIITΔ90 may have increased the abundance of functional basigin-MCT heterodimers (59, 98). Ultimately, to properly describe the physiological relevance of changes in acid-base transport protein content requires direct measurement of transport activity in vivo, or at least surrogate measurement of changes in substrate production and removal.

β_m_{in vitro} decreased slightly for both groups following training, although the reduction was less than our typical error of measurement (10.7 mmol H⁺·kg dm⁻¹·pH⁻¹). Contrary to our hypothesis, the failure to elicit increases in β_m_{in vitro} following two HIIT protocols with distinct energetic costs suggests that training intensity above the LT does not influence adaptations. This is unlikely to be related to the volume of training performed in the present study, as improved β_m_{in vitro} has previously been reported following similar or lower volumes of HIIT (14, 26, 37, 95). It may be that the titration technique is not sufficiently sensitive to detect the changes in β_m_{in vitro} typically found following training. In addition to the within-sample SD for repeat titrations in the current study of 10.7 mmol H⁺·kg dm⁻¹·pH⁻¹, our laboratory recently reported a within-subject, between-sample SD of 15.2 mmol H⁺·kg dm⁻¹·pH⁻¹ (70). However, the grand mean change reported to date for β_m_{in vitro} following training is only 9.2 mmol H⁺·kg dm⁻¹·pH⁻¹ (90% CL ± 5.0 mmol H⁺·kg dm⁻¹·pH⁻¹) (70), suggesting some reported changes are more indicative of methodological between-sample variability than physiological improvements in actual β_m. This places a question mark over conclusions made on the efficacy of different training modalities for inducing genuine changes in β_m. It also explains why β-alanine supplementation is unlikely to produce measurable changes in β_m_{in vitro}, given, for example, a reported 24% increase in carnosine content (38) would have resulted in an increase in β_m of only 3.3 mmol H⁺·kg dm⁻¹·pH⁻¹ (calculated from the Henderson-Hasselbalch equation, assuming a p*K_a* of 6.83). Ultimately, our data remain inconclusive as to whether training intensity influences β_m.

There were modest improvements in RSA after 4 wk of HIIT, with no difference between groups. While participants did show greater fatigue resistance during the RSA test, they did not perform more total work following 4 wk of HIIT. The latter sprints of a similar RSA test have previously been shown to require a greater O₂ uptake than the first sprint, with oxygen consumption approaching V_{O_{2max}}, during the final sprint (69). Therefore, the failure to improve V_{O_{2peak}} may have limited the capacity to sustain greater power for sequential sprints (69), whereas greater total work probably relates to an enhanced anaerobic capacity, resulting from improved PCr resynthesis (15). Furthermore, previous studies showing improvements in

RSA following similar HIIT interventions have failed to control for the effects of cadence on power. Without fixing cadence, it is not possible to discriminate changes in power output (or work), resulting from training adaptations, from the variation in maximal power dependent on the power-cadence relationship (35). In the present study, an isokinetic protocol with cadence fixed at 115 rpm was employed to quantify changes in power and work due to training, independent of cadence. Therefore, the modest training adaptations achieved here may more accurately represent the efficacy of this type of HIIT in improving RSA for an active male population. Further research is required to ascertain whether improvements in RSA can be achieved with just 4 wk of HIIT.

There was no change in mean $\dot{V}O_{2\text{peak}}$ for either group following training. This is in contrast to the grand mean effect of HIIT on $\dot{V}O_{2\text{max}}$ of 0.51 l/min (95% CL $\pm$ 0.08 l/min) reported in a meta-analysis of similar populations (3a). The meta-analysis did not include interventions of <4 wk, but did note that studies with the lowest effect were typically of shorter duration. This is consistent with an earlier study demonstrating a linear increase in $\dot{V}O_{2\text{max}}$ as the volume of training increased (46). In the present study, it is likely that 12 sessions of HIIT over 4 wk was an insufficient volume of training to allow the primarily central adaptations necessary to improve $\dot{V}O_{2\text{peak}}$ in active male participants. Similarly, both groups demonstrated a lack of improvement in W_{peak} , although there were apparent differences in the individual responses, with mean changes of 8 W (90% CL $\pm$ 7 W) and 8 W (90% CL $\pm$ 11 W) for HIIT Δ 20 and HIIT Δ 90, respectively. In contrast to W_{peak} , the greater duration of training performed at an intensity just above the LT by HIIT Δ 20 induced a mean increase in power at the LT of 22 W (90% CL $\pm$ 13 W), whereas there was no such mean change for HIIT Δ 90 (1 W; $\pm$ 11 W), which trained close to W_{peak} for a shorter duration. However, our group has previously found improvements in both the LT and W_{peak} following 5 wk of HIIT at a similar intensity to the HIIT Δ 90 group (29), but performing a 30% greater duration of training compared with HIIT Δ 90 (242 vs. 170 min total exercise duration). Despite the groups in the present study being work matched, it is probable the total exercise duration above the LT for HIIT Δ 20 (225 min) was more important in eliciting improvements in power at the LT than the relative intensity of training.

Consistent with previous research (91), we observed that an ostensibly low- or nonresponse to training held for some, but not all, measured variables in the same individuals. Part of the individual variability may relate to baseline acid-base protein content, and, although a larger sample size would be required to robustly correlate the data, for some proteins, namely, NHE1, CAII, and CAXIV, those with the highest preintervention protein content had the lowest response to training. And, interestingly, the highest responders for one protein were typically the highest responders for other proteins. There was also high apparent variability in the individual response for measures of aerobic capacity, although knowledge of the within-subject SD would be required to properly interrogate the individual training response (3). While it is probable there are, as yet, unidentified genetic factors explaining some of the training response (65), an additional possibility is that adaptations to training are random, owing to biological and technical variations (3). Notwithstanding repetition of an in-

tervention using the same participants after a sufficient wash-out period, robust study design and data analyses may help answer some of these questions in future (3, 45).

The present data support previous findings of decreased MCT1 and MCT4 abundance 6 wk after stopping training (20), and add to these data by showing rapid reversal of adaptations in basigin, NHE1, CAII, and CAXIV. It seems that maintaining adaptations in acid-base transport proteins requires the inclusion of some modality of high-intensity training. Similarly, detraining was apparent from a loss of RSA at +10 wk. By showing that adaptations can be rapidly induced with short-term HIIT, but equally can be rapidly lost, these data have important implications for periodization of high-intensity training. Determining the minimum "dose" of HIIT to maintain adaptations may provide practical information in the future.

In conclusion, there was increased abundance of basigin, MCT1, MCT4, NHE1, NBCe1, CAII, and CAXIV after 4 wk of HIIT, but, in general, six sessions over the first 2 wk provided insufficient stimulus to induce meaningful changes. Contrary to our hypothesis, an $\sim 40\%$ difference in training intensity did not influence adaptations for most of the measured proteins, apart from basigin and MCT4. The failure of either group to demonstrate improvements in $\beta_{\text{m in vitro}}$ suggests the intensity at which HIIT is performed above the LT is not critical. Finally, the present study found only modest improvements in RSA and aerobic capacity following 4 wk of HIIT. Others have previously reported improvements in exercise capacity without parallel increases in H^+ transport protein abundance (5, 40, 90). It is clear from the present study that the converse is also true: increased abundance of acid-base transport proteins can be shown without any change in performance.

ACKNOWLEDGMENTS

We are very grateful to the participants for their effort and commitment to this study. We also thank Dr. Mitch Anderson and Dr. Andrew Garnham for performing the muscle biopsies.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

C.M. and D.J.B. conception and design of research; C.M. performed experiments; C.M. analyzed data; C.M. and D.J.B. interpreted results of experiments; C.M. prepared figures; C.M. drafted manuscript; C.M. and D.J.B. edited and revised manuscript; C.M. and D.J.B. approved final version of manuscript.

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